Dermoscopic image segmentation method based on convolutional neural networks

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Abstract: In this paper, we present an efficient dermoscopic image segmentation method based on the linearisation of gamma-correction, and convolutional neural networks. Linearisation of gamma-correction is helpful to enhance low-intensity regions of skin lesion areas. Therefore, postprocessing tasks can work more effectively. The proposed convolutional neural network architecture for the segmentation method is based on the VGG-19 network. The acquired training results are convenient to apply the semantic segmentation method. Experimental results are conducted on the public ISIC-2017 dataset. To assess the quality of obtained segmentations, we make use of standard error metrics such as the Jaccard and Dice which are based on the overlap with ground truth, along with other measures such as the accuracy, sensitivity, and specificity. Moreover, we provide a comparison of our segmentation results with other similar methods. From experimental results, we infer that our method obtains excellent results in all the metrics and obtains competitive performance over other current and state of the art models for dermoscopic image segmentation.

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Keywords: dermoscopic images; deep CNNs; machine learning; skin lesions; image segmentation; skin cancer.

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1 Introduction

Cancer (Seyed and Mohammadamin, 2017) is a leading factor in diseases and causes a higher number of mortalities in the world. Until now, there are several methods to treat cancer, such as operation, chemotherapy, radiation therapy, immunotherapy and other modern combined techniques, such as targeted therapy. However, cancer treatment only achieves the best result if it is diagnosed early. If cancer goes to metastasis, the treatment is very

difficult, costs much, and usually, patients can only live for several years.

Cancer manifests in many human body parts and organs (Beddad et al., 2019; Chlebus et al., 2018; Bharati et al., 2020), such as the liver, lung, prostate, breast, stomach, blood, bone, mouth, etc. Skin cancer is also prevalent in the population and requires careful diagnosis and treatments.

Skin cancer happens mainly on people of Western countries because of sunbath habits and genetics. Among the different types of skin cancers, the most severe one is the melanoma since it grows very fast and causes the most death. According to the statistical data of the skin cancer association, just for USA alone, an estimated 192,310 cases were reported to be of melanoma in 2019. The main cause of skin cancer is to be exposed to the Ultra-Violet (UV) radiation. Because many people like outdoor activities under the sun without any skin protection, the number of cases of skin cancer continues to increase over time. These outdoor activities are gradually becoming popular for Asians. Hence, skin cancer is a serious and visible danger for all people.

In dermatology, skin cancer diagnosis begins from screening methods based on dermoscopic images. Doctors use modern technologies to analyse skin lesions and diagnose. One of the most popular and effective methods, in this case, is the ABCD rule (Kunz and Stolz, 2018). The ABCD rule uses several features of skin lesions, such as asymmetric features, border features, colours and diameter to evaluate a score for skin lesions (Thanh et al., 2020b; Moura et al., 2019; Nachbar et al., 1994). Imaging Diagnostic Systems (IDS) classify different forms of skin cancer by using the scores. Software integrated with IDS usually implements three stages to diagnose skin cancer: capture image and pre-process dermoscopic images, such as remove noise (Thanh et al., 2020a; Tian et al., 2020), remove hairs (Thanh et al., 2018b), extract ROI (regions of interests); segment the skin lesion; and extract various features of skin lesions to evaluate the score. In this work, we concentrate on the first and second stages, namely, the pre-processing and segmentation. The skin lesion segmentation is a very important task for skin cancer diagnosis because it is necessary to analyse deep features of skin lesions such as asymmetry, border features, colours and diameter by the ABCD rule.

Over the last two decades, various image processing models have been studied and applied for the skin lesion segmentation problem. They include the thresholding methods (Otsu, 1979; Ding et al., 2019; Thanh et al., 2019), the active contour methods (Thanh et al., 2018a; Zhang, 2019), and other learning-based methods (Yuan, 2017a; Yuan et al., 2017; Bi et al., 2017a; Berseth, 2017; Rashika and Ovidiu, 2017; Thanh et al., 2019). Learning-based methods are usually more effective than non-learning-based methods because they utilised the advantages of various learning models. Moreover, with many achievements of deep learning, we can design a network with multiple layers. Every layer can handle various tasks; and they can be assigned to a weight. By using machine learning methods, the weights can be estimated better than traditional methods.

For dermoscopic image segmentation, recently machine learning methods have become popular, within the particular focus on Deep Learning (DL) neural network models. Within a variety of DL models, the Convolutional Neural Networks (CNNs) have proven to be powerful in various biomedical image segmentation problems in general, and for skin lesions in particular. Yuan et al. (2017) proposed two skin lesion segmentation methods with fully CNNs with Jaccard distance, and with fully convolutional-deconvolutional networks (Yuan, 2017). Bi et al. (2017a) used deep residual networks to develop their own method. Berseth (2017) and Rashika and Ovidiu (2017) make use of the U-Net network for segmenting skin lesions on dermoscopic images. Recently, there are some novel learning-based methods for skin lesion segmentation such as the method based on high-resolution CNN (Xie et al., 2020), the method based on dense deconvolution network (Li et al., 2019), the method with deep convolution networks and unsupervised learning (Bozorgtabar et al., 2017) and the method based on multistage fully convolutional networks (Bi et al., 2017b). Accuracy and performance of the abovementioned methods are not high in the case that skin lesions have very low-intensity regions.

In this work, we study a dermoscopic image segmentation approach which is a combination of linearising gamma-correction and CNNs. Our contributions focus on enhancing low-intensity regions of skin lesion areas by the linearisation; proposing a CNN architecture for the semantic segmentation based on the VGG-19 network, and combining with the semantic segmentation method to implement the task of skin lesions segmentation. The proposed CNNs architecture was built based on a very deep network architecture that uses many layers; therefore, it may give a better accuracy compared to other similar methods with fewer layers. It is also combined with the linearisation of gamma correction to improve low-intensity regions of skin lesions to get a better result. Moreover, the proposed architecture is simple for modelling and implementing the training process.

We organised the work in the following way. Section 2 details our efficient deep CNN-based dermoscopic images segmentation approach. Section 3 provides detailed experiments on skin lesions images from the public ISIC-2017 data set and comparisons are given. Finally, Section 4 concludes the paper.

2 Dermoscopic image segmentation method

2.1 Gamma correction and linearisation

In computer graphics, the term "gamma correction" is used as a method for displaying an image accurately on a screen. The gamma correction controls the overall brightness/ contrast of an image (Ebner, 2007). With gamma correction, the image displayed on a screen will have a better white-balance. The image without being processed by gamma correction can look either bleached out or too dark. Today, almost all images being stored in sRGB (standard RGB) are processed by gamma correction. This standard defaults on a

vast amount of image capture systems. It is also used for dermoscopic images.

However, gamma correction is only better for human vision. In many processing tasks, it is necessary to handle original images without gamma correction. Hence, we need to linearise the gamma-corrected images. Linearisation of gamma-corrected images is a technique to undo gamma correction. There are two types of linearisation: by using sRGB and by using Adobe RGB. In this paper, we only consider sRGB model, because all dermoscopic images of the ISIC-2017 data set use this colour model. The sRGB tristimulus values are linearised by follows (Ebner, 2007):

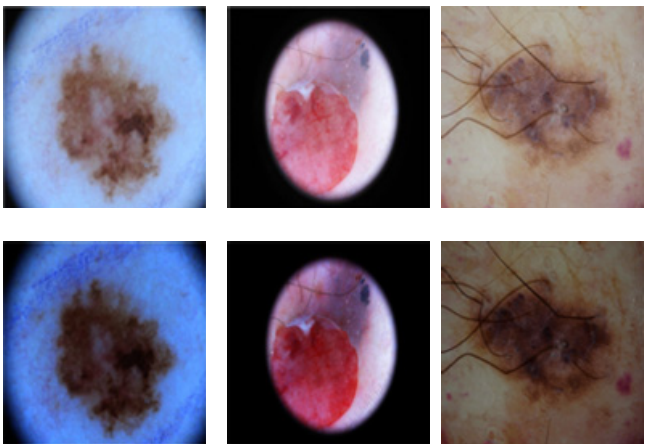
$$f(u) = \begin{cases} -f(u), if\ u < 0 \\ cu, if\ 0 \le u < d \\ (au+b)^{Y}, u \ge d \end{cases},$$

where

$$a = \frac{1}{1.055}; b = \frac{0.055}{1.055}; c = \frac{1}{12.92}; d = 0.04045; Y = 2.4.$$

Figure 1 shows an example of the linearisation of the gamma correction of dermoscopic images. From the visualisations, we can note that, after linearisation, regions with lower intensity of skin lesion areas get clearer. This is very useful for improving the accuracy of the segmentation methods considered here.

Figure 1 Linearizstion of gamma-corrected dermoscopic images: the first row – sRGB images; the second row – after linearization



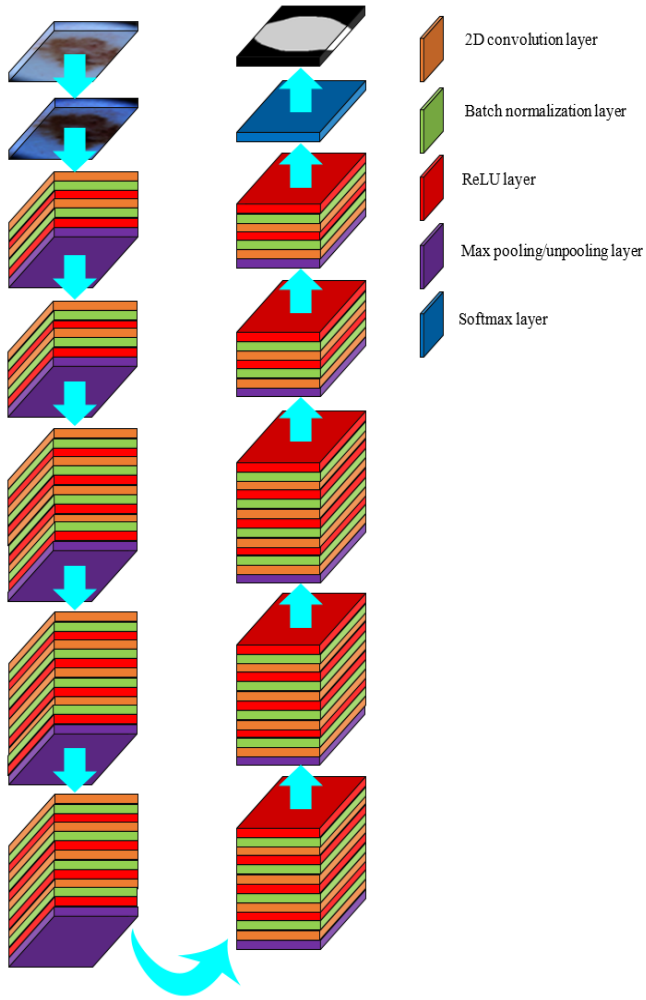
2.2 Proposed Convolutional Neural Networks Architecture

We design the convolutional neural network architecture for dermoscopic image segmentation base on the VGG-19 architecture (Krizhevsky et al., 2017). Figure 2 provides the overall architecture utilised here.

In the proposed CNN architecture, we use five main layers: the 2D convolution layer (LeCun et al., 1990), the batch normalisation layer (Ioffe and Szegedy, 2015), the ReLU layer (Vinod and Hinton, 2010), the max pooling/unpooling layer (Nagi et al., 2011) and the softmax layer (Bishop, 2006).

The main purpose of the 2D convolution layer is to apply the sliding convolution filters for the input image. The batch normalisation layer is used to normalise all input channels of the input image across a mini-batch. The ReLU layer corresponds to performing a threshold operation on each element of a given input image. The layer with max-pooling operation performs the down-sampling operator. In contrast, the layer with max unpooling unspools the output of the max-pooling layer. The softmax layer provides the application of the softmax function to the input image.

Figure 2 The proposed CNNs architecture for dermoscopic image segmentation



As we can see that the CNN architecture starts from an input image. After linearisation, it will be processed by pooling and unpooling operators in every block. We note that, in every block, if the violet-coloured layer is the end layer of a block (see Figure 2) and the arrow turns out from it, it is a max-pooling layer, otherwise, we define it as a max unpooling layer.

2.3 Proposed dermoscopic images segmentation method

We use the proposed CNNs architecture to train the segmentation model. Based on the trained CNNs, we combine it with the semantic segmentation method (Brostow et al., 2008) to implement the segmentation task.

Algorithm 1 presents the details of our dermoscopic images segmentation approach based on the linearisation of Gamma correction and CNNs. To implement Algorithm 1 in MATLAB, we use Image Processing, Computer Vision and Deep Learning toolboxes with the following functions:

- Function *rgb2lin(image_in_sRGB)*: linearise Gamma-correction of a standard RGB image.
- Function *semanticseg(image, a_trained_network)*: implement the semantic segmentation method.

Note that the Deep Learning toolbox is used for training the proposed CNNs model by using a function *trainNetwork*.

Algorithm 1: The proposed dermoscopic images-based skin lesion segmentation model

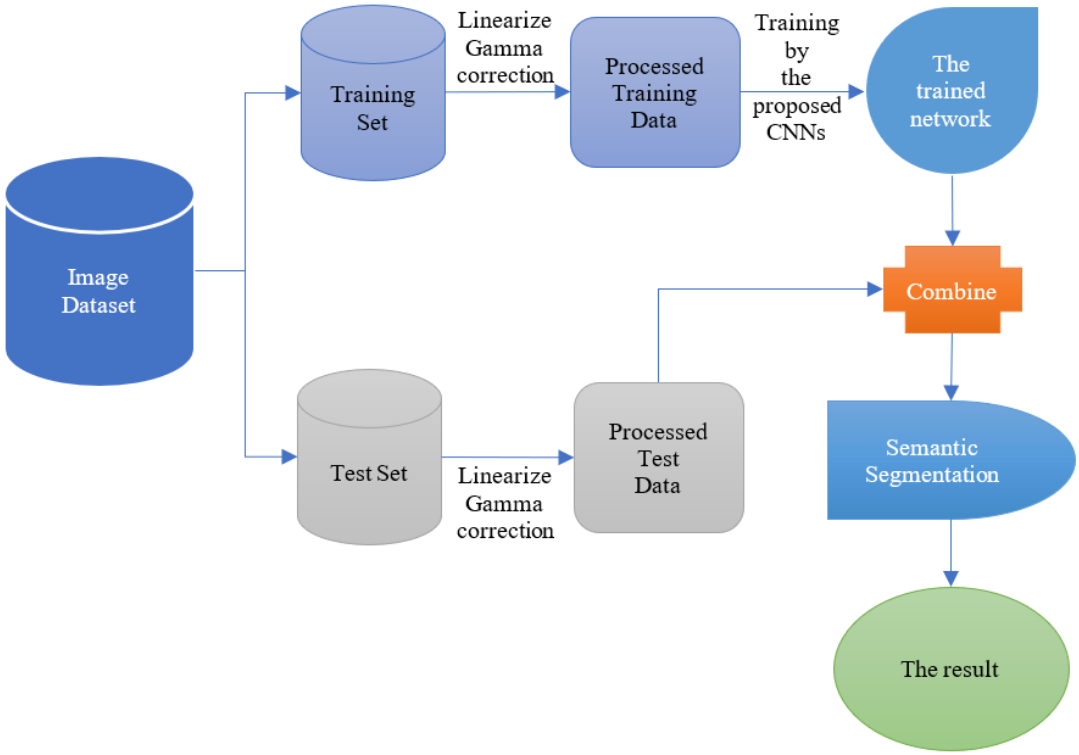
Begin

- Divide the image dataset into a training set and a test set.
- Linearise Gamma correction for dermoscopic images of both training and test sets.
- Train the image segmentation model by using the proposed CNNs architecture for the training set after linearising Gamma correction.
- Combine the semantic image segmentation method with the trained network:
  - If we want to assess the accuracy and performance of the segmentation task: segment images of the test set after linearising Gamma correction and evaluate scores.
  - If we just want to segment an input image: segment the image to obtain the binary segmented image.

End

The flowchart of the proposed skin lesion segmentation is presented in Figure 3.

Figure 3 Flowchart of the skin lesion segmentation method



3 Experimental results and discussions

In this work, we have implemented the proposed deep CNN architecture for skin lesion segmentation with the linearisation of gamma correction on a MATLAB 2018b computing environment.

3.1 Quality and error assessment metrics for quantitative comparison of segmentations

We utilise the following popular metrics for assessing the quality of the skin lesion segmentation tasks, namely Dice, and Jaccard. We further report on the Accuracy, Sensitivity (Recall), and Specificity. Note that these are well-known segmentation quality metrics and can summarise as follows (Gabriela et al., 2013; Abdel and Allan, 2015):

- *Accuracy, sensitivity and specificity metrics*:

Accuracy (%) assess how well a method correctly identifies a segmented region with a given ground truth:

$$accuracy = \frac{TP + TN}{FN + FP + TP + TN},$$

where TP, TN, FP, FN are true positive, true negative, false positive and false negative, respectively. These values are basic statistical indices that we can compute them by utilising predicted segmentation results and the given ground truth. We also remark that the ground truth segmentations were made manually by dermatologists with experience.

Sensitivity (%) assess the ratio of actual positives, that is also exactly identified:

$$sensitivity = \frac{TP}{FN + TP}.$$

Specificity (%) assess the ratio of actual negatives that are exactly identified:

$$specificity = \frac{TN}{FP + TN}.$$

- *The Dice similarity (F1-score)*: The Sørensen-Dice similarity (or Dice similarity, Dice score) (Abdel and Allan, 2015) is a better way to compare with the Accuracy. It is used for measuring accuracy of segmentation result and it is evaluated as follows:

$$dice = \frac{2TP}{FP + FN + 2TP}.$$

The Sørensen-Dice similarity metric provides a value between 0 and 1, with higher values indicating better segmentation results.

- *The Jaccard similarity*: This metric, known as the Jaccard metric (or Jaccard score) (Abdel and Allan, 2015) has a close relationship to the classical Dice metric defined above and is given as:

$$jaccard = \frac{dice}{2 - dice}.$$

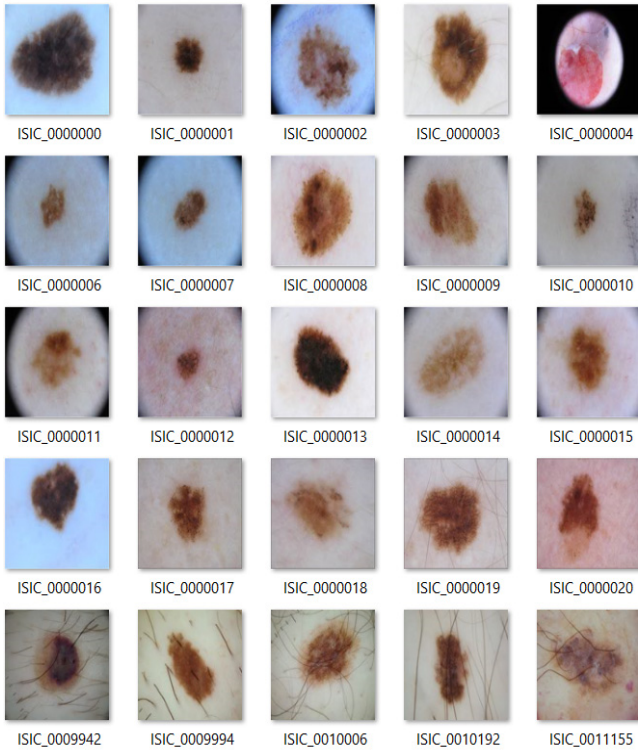
The Jaccard value is in the range of 0% to 100% with higher values indicating better segmentation results.

We further note that according to the literature on medical image segmentation, the most important value to track for image segmentation quality is the Jaccard metric, and its value lays in the range of [0–1] or [0–100%]. The higher value indicates a better segmentation quality from the method.

3.2 Dermoscopic image data sets

We have used dermoscopic images of the ISIC-2017 data set containing 2000 colourful images that can be found at *https://challenge.kitware.com/#phase/584b0afacad3a51cc66c8e24*. All images are in the RGB colour space in JPEG format. We resize the original images to 128 × 128 pixels to increase the processing speed. Several examples of the tested images are shown in Figure 4.

Figure 4 Several examples of dermoscopic images from the public ISIC-2017 data set



3.3 Experiments with deep CNN and discussion

In the training process, we configure parameters as follows: the training method is the Stochastic Gradient Descent (SGD) coupled with momentum. We also set the initial learning rate to 10^{-3}, the max epoch is 200, mini-batch size is 32. The training time is about 8 hours.

Figure 5 illustrates the results of the training process in terms of loss and accuracy (%). The mini-batch accuracy is 96.08%, the mini-batch loss is 0.3285 after 200 epochs.

It can be seen that in the case of epoch less than 100, the accuracy increases very fast and the loss decreases fast too. From the epoch point of 100 forward, the change of the

accuracy and the loss are slower. Overall, the final accuracy of the training process is very good.

After training, we implement the semantic segmentation method. We choose 25 dermoscopic images randomly to implement the segmentation procedure and evaluate the scores. The average scores are used for comparing with other similar skin lesion segmentation methods.

In Figure 6, the first column is for original images of the ISIC-2017 data set; the second column is for binary segmentation results; and the third column shows that the segmented boundaries of the proposed method, along with the ground truth in red and green borders, respectively. It can be seen that the difference of the boundaries of the skin lesion area segmented by the proposed method and by the ground truth is just a little.

Figure 5 Mini-batch Accuracy (96.08%) and Mini-batch Loss (0.3285) of the training process, with 200 epochs

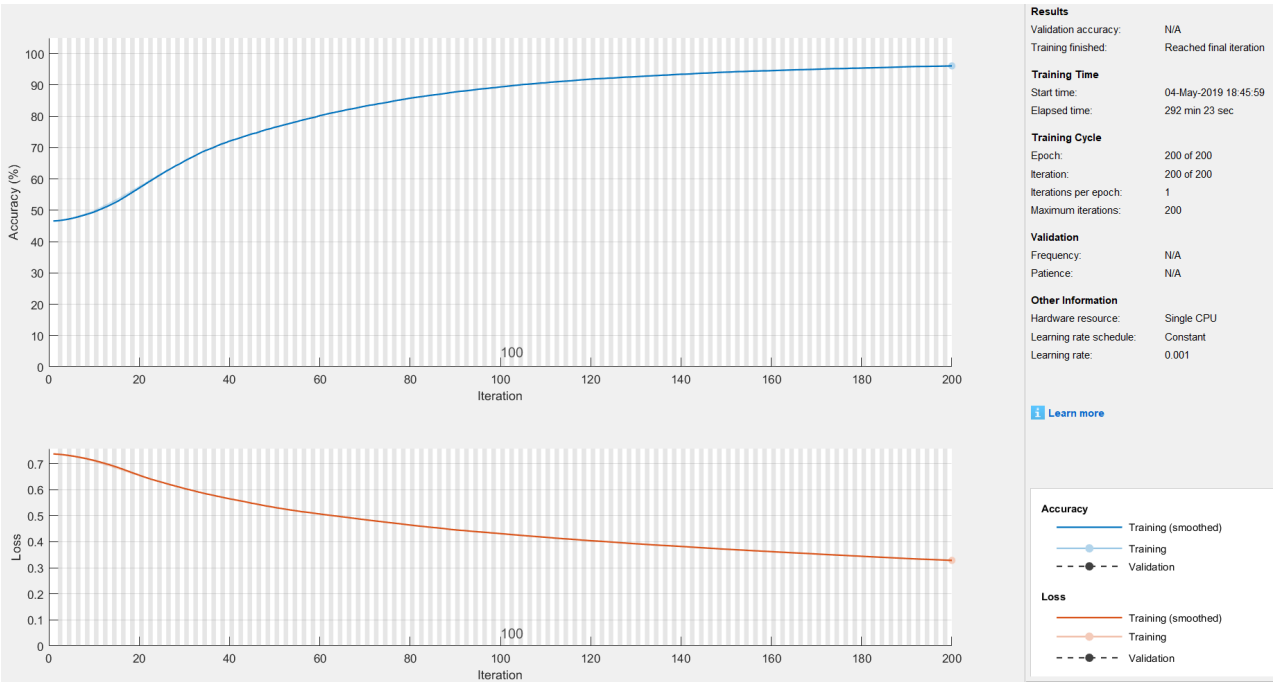


Figure 6 Skin lesion segmentation results. Left column – original images, middle column – binary segmented images and right column – boundaries of segmented areas of our proposed approach and the ground truth (the red border – for our method, the green border – for the ground truth)

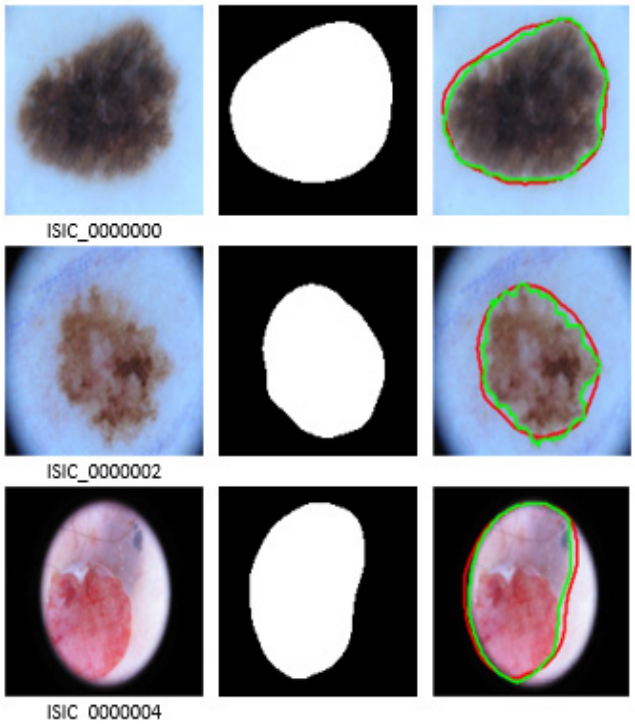


Figure 6 Skin lesion segmentation results. Left column – original images, middle column – binary segmented images and right column – boundaries of segmented areas of our proposed approach and the ground truth (the red border – for our method, the green border – for the ground truth) (continued)

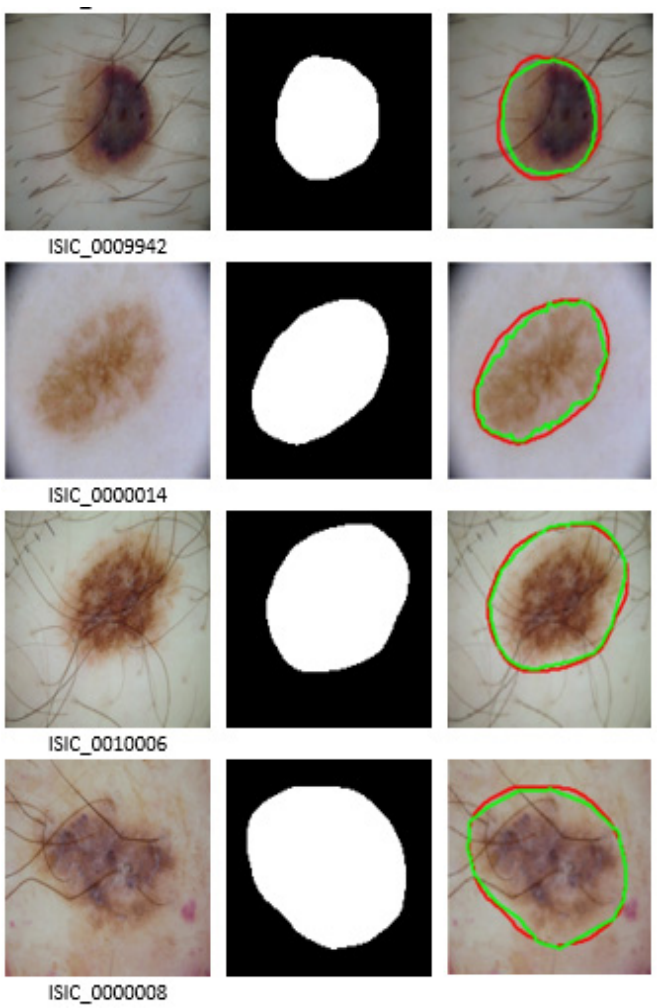


Table 1 lists the error metrics Dice and the Jaccard, along with the accuracy, sensitivity, specificity scores of the proposed method for every selected image, and the corresponding average scores. The acquired scores are very impressive. The highest Dice metric obtained is 96.5%, whereas the highest Jaccard value obtained is 93.3%. The highest values of accuracy sensitivity, specificity scores are 98.4%, 100% and 98.3%, respectively. In some cases, the sensitivity score reaches the maximum value (100%). The average scores are also high indicating the effectiveness of the model across images in the database.

In Table 2, we can infer that the comparison of the average score of the proposed deep CNN-based method along with other similar skin lesion segmentation models from the literature. All the considered segmentation methods here are based on DL-based modes. Dice, Jaccard, accuracy and sensitivity scores of our method are the highest. The segmentation result of the method of Rashika and Ovidiu (2017) also achieves the same average Jaccard score with our method. However, the segmentation results of our proposed method are better, when compared to Rashika and Ovidiu method (2017) for the remaining scores including important scores, such as Dice, accuracy scores. Further, for the specificity score, the segmentation results with our proposed method are the only ones higher than the Rashika and Ovidiu method (Rashika and Ovidiu, 2017). However, Jaccard, Dice and accuracy scores are more important, since it is known that Jaccard (thereby closely related Dice) values are considered as the most important metrics for medical image segmentation (Yuan et al., 2017; Gabriela et al., 2013).

Table 1 The scores of dermoscopic image segmentation quality for the proposed method

Images	*Dice*	*Jaccard*	*Accuracy*	*Sensitivity*	*Specificity*
ISIC_0000000	0.965	0.933	0.967	0.981	0.955
ISIC_0000001	0.821	0.697	0.970	0.998	0.968
ISIC_0000002	0.931	0.870	0.956	0.984	0.943
ISIC_0000003	0.949	0.903	0.965	0.991	0.952
ISIC_0000004	0.949	0.903	0.963	0.992	0.948
ISIC_0000006	0.862	0.757	0.972	0.988	0.970
ISIC_0000007	0.860	0.754	0.972	1.000	0.969
ISIC_0000008	0.947	0.898	0.963	0.988	0.950
ISIC_0000009	0.919	0.851	0.960	0.998	0.949
ISIC_0000010	0.846	0.733	0.972	0.969	0.972
ISIC_0000011	0.904	0.824	0.962	0.999	0.954
ISIC_0000012	0.859	0.753	0.984	1.000	0.983
ISIC_0000013	0.937	0.882	0.967	0.998	0.957
ISIC_0000014	0.931	0.870	0.959	1.000	0.943
ISIC_0000015	0.906	0.828	0.951	0.996	0.937
ISIC_0000016	0.921	0.853	0.965	0.987	0.959
ISIC_0000017	0.882	0.789	0.959	1.000	0.952
ISIC_0000018	0.925	0.860	0.962	0.994	0.953
ISIC_0000019	0.958	0.919	0.972	0.990	0.964
ISIC_0000020	0.899	0.817	0.949	0.973	0.942
ISIC_0009942	0.921	0.853	0.966	1.000	0.958
ISIC_0009994	0.876	0.780	0.955	0.974	0.951
ISIC_0010006	0.963	0.928	0.974	0.989	0.966
ISIC_0010192	0.921	0.853	0.965	0.987	0.959
ISIC_0011155	0.964	0.930	0.968	0.995	0.949
AVERAGE	0.913	0.842	0.965	0.991	0.956

Table 2 The scores comparison of the dermoscopic images-based skin lesions segmentation methods

Methods	*Accuracy*	*Dice*	*Jaccard*	*Sensitivity*	*Specificity*
Bi et al. (2017a)	0.934	0.844	0.760	0.802	**0.985**
Rashika and Ovidiu (2017)	0.928	0.868	**0.842**	0.930	0.842
Yuan (2017)	0.934	0.849	0.765	0.825	0.975
Berseth (2017)	0.932	0.847	0.762	0.820	0.978
Proposed without Linearisation of Gamma correction	0.934	0.875	0.778	0.965	0.924
Proposed with Linearisation of Gamma correction	**0.965**	**0.913**	**0.842**	**0.991**	0.956

4 Conclusions

In this article, we have studied a dermoscopic image segmentation method driven by an efficient deep Convolutional Neural Network (CNN) model from the Deep Learning (DL) area. The method is developed by utilising the linearisation of gamma-correction and leverages deep CNN for obtaining a better skin lesion segmentation result. The architecture of our proposed deep CNN is based on the VGG-19 network and proven to be consistent with the semantic segmentations. Through experimental results on various skin lesions images, we showed that the proposed DL model-based approach works very effectively. Further, we obtained better results quantitatively than other related deep learning models for skin lesion segmentation from the literature. Other deep learning models that utilise deeper layers can also be utilised to improve the accuracy further and defines our future works in this direction.

In the future, we will implement the tests for the proposed method on various datasets of skin lesion images. Although ISIC is the largest data set of skin lesions, testing on various datasets is helpful to improve the accuracy and performance of the proposed method.

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